Yichen Han

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Trustworthy and interpretable statistical learning for genomic biomarker discovery

Education

M.Sc. Statistics & Data Science Sep 2027 (expected)
Ludwig-Maximilians-Universität München

Machine Learning track.

B.Sc. Statistics & Data Science Oct 2022 – Jan 2026
Ludwig-Maximilians-Universität München Grade: 1.25 (German scale; 1 = best)

Minor in Biology (30 ECTS)

Thesis: *Robustness-Driven Feature Attribution in Genomic Deep Learning with Integrated Gradients*

Investigated when adversarial robustness improves interpretation plausibility in genomic CNNs. Supervised by Bernd Bischl and Martin Binder. Grade: 1.0.

Abstract · Slides · GitHub

Publications

Münch P.C., Safaei N., Mreches R., Binder M., **Han Y.**, Robertson G., Franzosa E.A., Huttenhower C.*, McHardy A.C.* *Comparative Assessment of Large Language Models for Microbial Phenotype Annotation.* Revise & Resubmit at *Genome Biology*, 2025. bioRxiv.
(Contribution: statistical analysis, ensemble learning methods)

Aleksic M., Ehring T., Kunze A., **Han Y.**, Funk H., Wolkenstein L.* *Selective Effects of EMDR, Imagery Rescripting and Imaginal Exposure on Voluntary and Involuntary Memory of an Aversive Autobiographical Event.* Revise & Resubmit at *Journal of Experimental Psychology: Applied*, 2025. OSF.
(Contribution: statistical guidance, methods & results review)

Lin L., Hatzikotoulas K., …, Wiegrebe S., **Han Y.**, Ivanova O., …, Zeggini E., Rachow A.*, Held K.* *Genetic Insights into Lung Function and Its Trajectory among People with Tuberculosis in Africa: A Genome-Wide Association Study.* Under review at *Nature Communications*, 2026.
(Contribution: statistical guidance, methods & results review)

*Senior/corresponding author

Research Experience

The Huttenhower Lab, Harvard T.H. Chan School of Public Health Jan 2025 – present
Collaborating Student Researcher (joint with Philipp Münch) Boston, MA, USA (remote)

- Leading the interpretable genomic deep learning project, advised by Curtis Huttenhower
- Designed synthetic genome datasets for interpretability benchmarking, identifying compositional biases as key confounders
- Developed evaluation framework for assessing biological plausibility of model explanations
- Presented project findings at the departmental research group seminar (Aug 2025, May 2026)

BIFO Group, Helmholtz Centre for Infection Research Oct 2024 – present
Research Assistant (PI: Alice McHardy; Team lead: Philipp Münch) Braunschweig, Germany (remote)

- Developed adversarial training pipelines for genomic CNNs (PyTorch)
- Implemented Integrated Gradients for genomic models (Python, R)
- Developed statistical tools for Rashomon effect evaluation, revealing interpretation instability in real data
- Designed and implemented calibrated models for BacDive phenotype prediction (R) GitHub

Statistical Consulting Unit (StaBLab), LMU Munich — Apr 2024 – Jun 2026
Research Assistant & Statistical Consultant — Munich, Germany

- Methodological support to 30+ graduate projects (MSc/PhD) in biology, medicine, psychology
- Methods: GLMMs/GAMs, Bayesian methods, imputation, spatial-temporal, causal DAGs, applied DL
- Contributed to the BfS-funded Wismut project: Bayesian hierarchical modeling; Model evaluation and HPC pipeline development.

Academic Projects (selected)

The Misspecification Bias in TSLS with Covariates — Oct 2024 – Mar 2025
Causal Inference Seminar (Master's level), Grade: 1.0, GitHub — LMU Munich

- Supervised by Daniel Wilhelm.
- Investigated conditions under which Two-Stage Least Squares (TSLS) with covariates yields valid causal interpretations, synthesizing recent results on "rich covariates" for positively weighted treatment effects
- Demonstrated misspecification bias through simulations; surveyed uptake in subsequent literature

Wildlife abundance modeling — Feb 2025 – May 2025
Advanced Statistical Practicum, Grade: 1.0 — LMU Munich

- Evaluated N-Mixture models for estimating wildlife abundance from repeated camera-trap data across ten German national parks (collaboration with Nationalparkverwaltung Bayerischer Wald)
- Wrote a review on N-Mixture model applications and theoretical gaps

Teaching Experience (selected)

Teaching Assistant — Oct 2023 – Mar 2026
Department of Statistics — LMU Munich

- **Tutorial Instructor** – Descriptive Statistics & EDA (Winter 2023/24): weekly tutorials for $\sim$120 students; designed exercises
- **Project Supervisor** – Advanced Statistical Practicum (Winter 2025/26): supervised 3 student projects (Greek tomb kinship modeling; stroke metabolomics; TCGA-HNSCC differential expression analysis)
- **Curriculum Developer** – Descriptive Statistics & Probability Theory (Summer 2024): primary author of exam-prep materials; workshops for $\sim$80 students

Service & Leadership

Elected Member, Faculty Council (*Fakultätskonvent*) — Jul 2025 – Jul 2027
Faculty of Mathematics, Informatics and Statistics — LMU Munich

Student Representative — Mar 2025 – present
Two W2 Professorship Appointment Committees — Department of Statistics, LMU Munich

Student Council Statistics & Data Science — Apr 2023 – present
Elected Member (two terms, Jul 2025 – Jul 2027); Co-organizer, Round Table for Teaching — LMU Munich

Skills

Programming	R · Python · SAS · Bash
Tools	SLURM, Snakemake, Docker, SQL, Git, Linux, LaTeX, Inkscape
Languages	Chinese (native), German (C1), English (IELTS 8.0)